

On Calculations of Legendre Functions and Associated Legendre Functions of the First Kind of Complex Degree

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Abstract Formulas for calculating Legendre functions and associated Legendre functions of the first kind of complex degree using an Ermakov–Lewis invariant are presented. These formulas are straight-forward to implement numerically and are motivated by the lack of computational routines in standard university tools like those of MatLab and Maple. Angular waves propagating in opposite directions are also obtained. The results are particularly useful in complex angular momentum theories and nearside/farside analysis of spin-dependent angular scattering from central potentials.

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1 Introduction

The basic complex angular momentum (CAM) representations of differential cross sections involve Legendre functions of complex degree related to Regge poles of the S -matrix and complementary background integrals.^[1–4] In heavy-particle scattering the important angular momenta for analyzing differential cross sections are usually large ($|l| \gg 1$), with a dominating positive real part, and reliable asymptotic techniques can be used for computations of the Legendre functions. Some problems may occur with less sophisticated (non-uniform) techniques near forward ($\theta = 0$) and backward ($\theta = \pi$) angles. In electron-atom scattering the use of CAM theory^[1,2,4] sometimes requires Legendre functions (and associated Legendre functions) of complex degree of the order of unity where asymptotic approximations of the Legendre functions are not reliable. Algorithms for these functions are not routinely implemented in common packages, the results of the present study will be particularly useful in research involving complex angular momenta.

Various approximate methods for calculating Legendre functions have been proposed; see Connor and Mackay 1978, 1979^[3] and references cited there. In the present work, the interest is only on exact numerical computations of Legendre functions with a complex degree. The suggested numerical approach allows all Legendre functions to be computed in terms of a non-oscillating function and is easy to implement in e.g. *MatLab*. The special non-oscillating function and the Legendre function of interest satisfy a so-called Ermakov–Lewis invariant.^[5,9–15] Therefore such calculations are very effective in a numerical sense.

Section 2 reviews the relevant equations for associated Legendre functions of the first kind. The Ermakov–Lewis invariant is introduced in Sec. 3. Numerical applications

for integer and complex degrees are given in Sec. 4. Section 5 contains conclusions.

2 Legendre Functions of the First Kind

The Legendre functions $P_l^m(\cos \theta)$, with complex degree l and $m = 0, 1, \dots$, satisfy the differential equation^[16]

$$y'' + \cot \theta y' + \left(l(l+1) - \frac{m^2}{\sin^2 \theta} \right) y = 0, \quad (1)$$

where the prime ' denotes differentiation with respect to the angle θ ($0 \leq \theta \leq \pi$). No particular distinction between Legendre ($m = 0$) or associated Legendre ($m = 1, 2, \dots$) functions is needed in this and the following sections, which allows a simplified notation of auxiliary quantities.

By introducing a new angular function

$$\chi = (\sin \theta)^{1/2} y, \quad (2)$$

in Eq. (1), its defining differential equation becomes

$$\chi'' + \left((l+1/2)^2 + \frac{1-4m^2}{4\sin^2 \theta} \right) \chi = 0. \quad (3)$$

Values for χ and χ' relevant for computing $P_l^m(\cos \theta)$ are

$$\chi_{\pi/2} = \frac{2^m}{\sqrt{\pi}} \cos \left(\frac{\pi(l+m)}{2} \right) \frac{\Gamma(1/2 + (l+m)/2)}{\Gamma(1 + (l-m)/2)}, \quad (4)$$

$$\chi'_{\pi/2} = -\frac{2^{m+1}}{\sqrt{\pi}} \sin \left(\frac{\pi(l+m)}{2} \right) \frac{\Gamma(1 + (l+m)/2)}{\Gamma(1/2 + (l-m)/2)}, \quad (5)$$

which are given in Ref. [17]. The notation here is such that $\chi_{\pi/2}$ indicates the value $\chi(\pi/2)$ and similarly for its derivative. The special values in Eqs. (4) and (5) are valid for complex values of the degree l .

3 Ermakov–Lewis Invariant

The Ermakov–Lewis invariant combines two functions, and their derivatives.^[12] The function χ in an Ermakov system has a partner function u satisfying a differential

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equation given by

$$u'' + \left((l + 1/2)^2 + \frac{1 - 4m^2}{4 \sin^2 \theta} \right) u = u^{-3}, \quad (6)$$

often called a Milne–Pinney type differential equation. The invariant constant of motion is^[9–11]

$$\mathcal{L} = \frac{1}{2} \left[(\chi' u - \chi u')^2 + \left(\frac{\chi}{u} \right)^2 \right], \quad (7)$$

which can be verified by direct differentiation of Eq. (7) and using the differential equations for χ and u .

Based on the quadratic invariant (7), one can write the two variable components in the invariant as:

$$\frac{\chi}{u} = \sqrt{2\mathcal{L}} \sin \beta, \quad (8)$$

$$(\chi' u - \chi u') = \sqrt{2\mathcal{L}} \cos \beta, \quad (9)$$

where β is an unspecified phase function of the real angle θ . The derivative of Eq. (8) is

$$(\chi' u - \chi u')/u^2 = \sqrt{2\mathcal{L}} \beta' \cos \beta, \quad (10)$$

and making use of Eq. (9) one observes that the derivative of the phase β then has to satisfy

$$\beta' = \frac{1}{u^2}. \quad (11)$$

Equation (8) implies that $\chi = \sqrt{2\mathcal{L}} u \sin \beta$ is a representation of χ in terms of u , since β is determined from u apart from an arbitrary constant phase. By writing the phase as

$$\beta = \phi + C, \quad (12)$$

with

$$\phi = \int_{\pi/2}^{\theta} u^{-2}(t) dt, \quad (13)$$

being a well defined phase for a given function u and with C being an arbitrary constant, one can instead of $\sin \beta$ write $a \sin \phi + b \cos \phi$ with $|a|^2 + |b|^2 = 1$. Hence, a general representation of χ is given by

$$\chi(\theta) = \sqrt{2\mathcal{L}} u(\theta) (a \sin \phi(\theta) + b \cos \phi(\theta)). \quad (14)$$

This representation is exact for any amplitude function u computed from Eq. (6). The main issue is rather to use a solution u as simple as possible.

In Eq. (13), the angle $\theta = \pi/2$ is the phase reference point. At this point the integration of the “amplitude” function u is initiated with a suitable value denoted $u_{\pi/2}$. A suitable value here means any value that makes the calculated function u vary reasonably slowly in the angular interval $0 < \theta < \pi$. This is a particular feature of the present method; no exact initial conditions are needed.

The numerical integration made in the present calculations assumes that u' and u'' have negligible values at $\theta = \pi/2$. Hence, a leading-order adiabatic solution is used here as the initial conditions for integrating u from Eq. (6). Milne’s differential equation then simplifies to

$$\left((l + 1/2)^2 + \frac{1 - 4m^2}{4 \sin^2 \theta} \right) u \approx u^{-3}. \quad (15)$$

This algebraic equation has an approximate solution

$$u \approx \left((l + 1/2)^2 + \frac{1 - 4m^2}{4 \sin^2 \theta} \right)^{-1/4}, \quad (16)$$

which is a symmetric function with respect to $\theta = \pi/2$. The initial conditions consistent with Eq. (16) are

$$u_{\pi/2} = \left((l + 1/2)^2 + \frac{1 - 4m^2}{4} \right)^{-1/4}, \quad u'_{\pi/2} = 0. \quad (17)$$

At this stage the function u is completely determined. However, as already pointed out, there may be other initial conditions that turn out to be more efficient for the amplitude-phase method.^[6]

Since the coefficient in Eq. (6) is symmetric with respect to $\theta = \pi/2$ and $u'_{\pi/2} = 0$, the initial conditions (17) imply the numerical function u has to have the same symmetry.

The invariant $\mathcal{L}$ is determined at $\theta = \pi/2$ from functional values of χ and u there, i.e.

$$\mathcal{L} = \frac{1}{2} \left[(\chi'_{\pi/2} u_{\pi/2})^2 + \left(\frac{\chi_{\pi/2}}{u_{\pi/2}} \right)^2 \right]. \quad (18)$$

It remains to determine the constants a and b in the representation (14). Hence, from Eq. (14) and its derivative, using Eqs. (4) and (5), one finds

$$\begin{pmatrix} \chi_{\pi/2} \\ \chi'_{\pi/2} \end{pmatrix} = (2\mathcal{L})^{1/2} \begin{pmatrix} 0 & u_{\pi/2} \\ u_{\pi/2}^{-1} & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}, \quad (19)$$

which implies

$$\begin{pmatrix} a \\ b \end{pmatrix} = (2\mathcal{L})^{-1/2} \begin{pmatrix} 0 & u_{\pi/2} \\ u_{\pi/2}^{-1} & 0 \end{pmatrix} \begin{pmatrix} \chi_{\pi/2} \\ \chi'_{\pi/2} \end{pmatrix}. \quad (20)$$

The function $\chi(\theta)$ is now well defined.

4 Legendre Functions of the First Kind

From Eqs. (2), (14), and (20) a Legendre function of the first kind is thus given by the exact representation

$$P_l^m(\cos \theta) = (\sin \theta)^{-1/2} u(\theta) (\chi'_{\pi/2} u_{\pi/2} \sin \phi(\theta) + \chi_{\pi/2} / u_{\pi/2} \cos \phi(\theta)), \quad (21)$$

where for real values of l one has a, b real, with u and ϕ real functions. The invariant $\mathcal{L}$ does not appear explicitly.

It is interesting to note from Eq. (21) the obvious decomposition of $P_l^m(\cos \theta)$ in terms of Jost-type propagating angular functions

$$P_l^{m(\pm)}(\cos \theta) = \frac{1}{2} (\sin \theta)^{-1/2} u(\theta) (\chi_{\pi/2} / u_{\pi/2} \mp i \chi'_{\pi/2} u_{\pi/2}) \exp(\pm i \phi(\theta)), \quad (22)$$

satisfying

$$P_l^m(\cos \theta) = P_l^{m(+)}(\cos \theta) + P_l^{m(-)}(\cos \theta). \quad (23)$$

The functions $P_l^{m(\pm)}(\cos \theta)$ are suitable for analyzing nearside/farside angular wave contributions in scattering problems; see Ref. [18].

In the representations (21) and (22) one can use in computations that $u(\theta)$ is in general complex and symmetric whilst $\phi(\theta)$ is complex and antisymmetric with respect to $\theta = \pi/2$.

5 Computations and Results

The basic function $u(\theta)$ calculated from $\theta = \pi/2$ towards $\theta = 0$, given the initial conditions in Eq. (17), has typical behaviors shown in Fig. 1 where $l = 10$ and $m = 0, 1$. When $m = 0$, $u(\theta)$ approaches zero as $\theta \rightarrow 0$,

whereas for $m \geq 1$ it increases indefinitely. In a large angular range near $\theta = \pi/2$, $u(\theta)$ is almost constant.

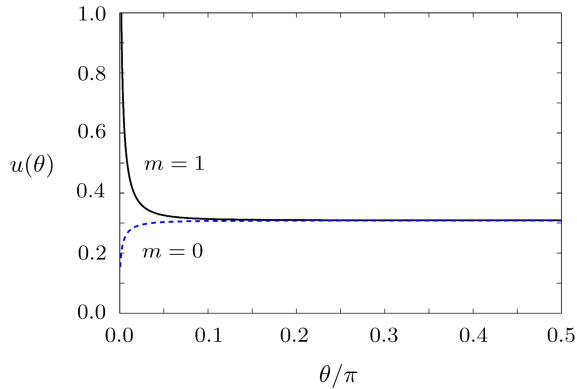


Fig. 1 The functions $u(\theta)$ for $l = 10$ and $m=0, 1$. For $m = 1$, $u(\theta)$ tends to infinity as $\theta \rightarrow 0$.

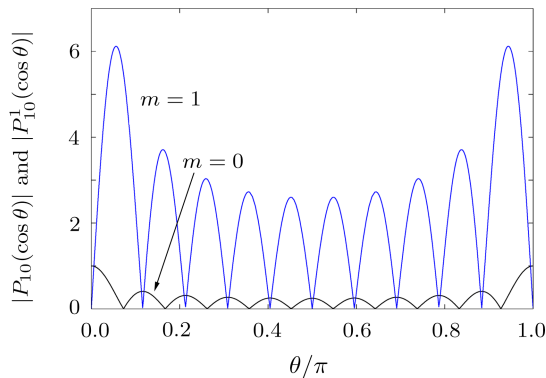


Fig. 2 Absolute values of Legendre polynomials with $(l, m)=(10,0)$ (black continuous line) and associated Legendre functions with $(l, m) = (10, 1)$ (blue continuous line).

From $u(\theta)$ and its initial conditions one then has the

representation of $P_l^m(\cos \theta)$ in Eq. (21). For integer values of l this representation confirms results from those obtained from well-known recurrence formulas.^[17] However, no comparisons with standard numerical tools could be made for complex values of l . The illustration in Fig. 2 shows that $|P_{10}^1(\cos \theta)|$ has larger oscillation amplitudes than $|P_{10}(\cos \theta)|$, and this is a general feature for associated Legendre polynomials due to their normalization.

Figures 3–4 show results from amplitude-phase calculations with complex degree $l = 4 + i0.5$. In Fig. 3, two subplots are shown, representing the real and imaginary parts, respectively, of the Legendre function with $m = 0$ (dashed line) and its propagating components (continuous lines). All curves in the left subplot are regular in the angular range. The propagating Legendre functions in the right subplot are singular at $\theta = 0, \pi$, which is an unphysical feature. At $\theta = 0$ the singular behaviors of the propagating Legendre functions cancel so that the imaginary part of $P_l(\cos \theta)$ becomes regular there; see Eq. (23). At larger angles both real and imaginary parts of $P_l(\cos \theta)$ approach those of $P_l^{(-)}(\cos \theta)$.

The propagating associated Legendre functions $P_l^{1(\pm)}(\cos \theta)$ (continuous lines) are analyzed and compared with $P_l^1(\cos \theta)$ (dashed lines) for $l = 4 + i0.5$ in Fig. 4. The left subplot shows the real parts of the functions. Again, all curves are regular in the angular range, and the real part of $P_l^1(\cos \theta)$ approaches the real part of $P_l^{1(-)}(\cos \theta)$ at larger angles. The right subplot shows the imaginary parts of the same functions. The propagating associated Legendre functions are singular at $\theta = 0, \pi$, but their singularities at $\theta = 0$ cancel when combined to $P_l^1(\cos \theta)$; see Eq. (23). The imaginary part of $P_l^1(\cos \theta)$ approaches the imaginary part of $P_l^{1(-)}(\cos \theta)$ at larger angles.

Obviously, special care is needed in interpreting a side/farside analysis of scattering cross sections near $\theta = 0, \pi$.

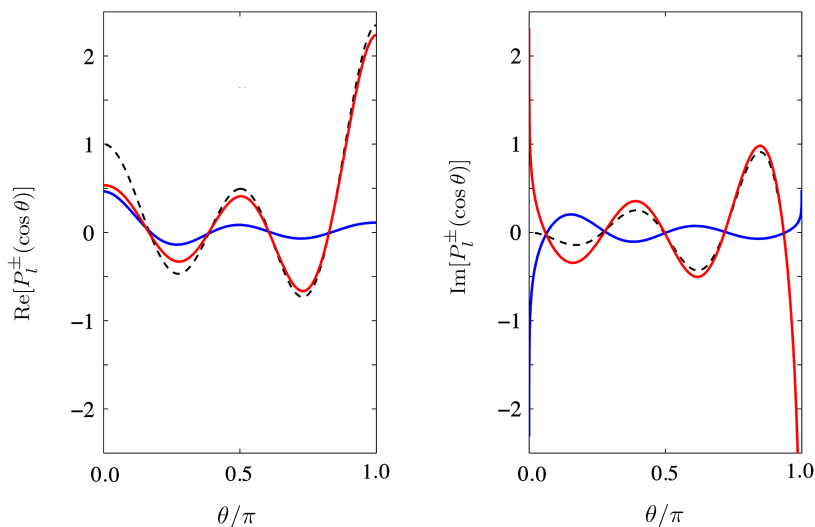


Fig. 3 Real and imaginary parts of the propagating Legendre functions with $(l, m) = (4.0 + 0.5i, 0)$ (continuous lines, (+)-waves are blue and (-)-waves red) in comparison with their sums, the complete Legendre function (dashed line).

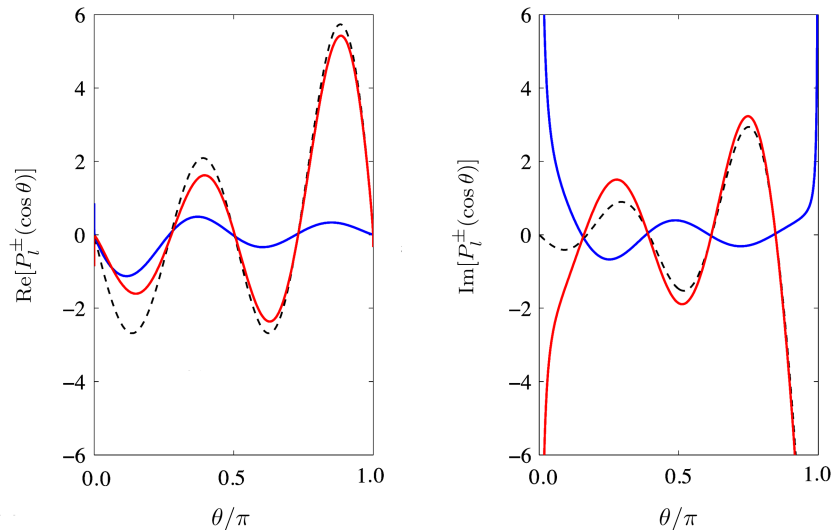


Fig. 4 Real and imaginary parts of the propagating associated Legendre functions with $(l, m) = (4.0 + 0.5i, 1)$ (continuous lines, (+)-waves are blue and (-)-waves red) in comparison with their sums, the complete associated Legendre function (dashed line).

6 Conclusions

The present results are relevant for computations of the Legendre function of complex-valued degrees. A representation of $P_l^m(\cos \theta)$ in terms of a symmetric and non-oscillatory function has been derived. It applies to arbitrary complex values of l . The method is primarily aimed at computing $P_l^m(\cos \theta)$ and its angular propagating components $P_l^{m(\pm)}(\cos \theta)$ for a limited number of arbitrary complex l -values.

A particular advantage of the present method (for integer or complex degrees) is the direct decomposition of Legendre functions into their nearside/farside contributions of differential cross sections.^[18] Results are also relevant for Regge-pole (CAM) analysis of differential cross sections of quantum particle scattering, and are valid for scattering angles near the limits $\theta = 0$ and π . The present results indicate that interpretations in terms of propagating waves need special care near forward and backward angles.

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